



BIT702 ASSESSMENT 2 SUPPORTING DOCUMENTATION

Web Technologies

Molly Smith

Contents

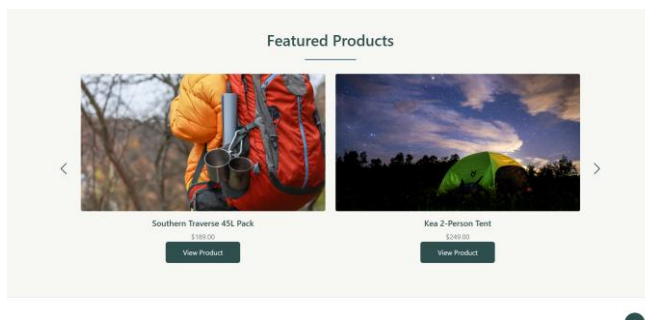
Task 2 - Use JavaScript to create a range of interactive elements.....	2
Task 2a - You must include at least four elements of functionality that are appropriate for an e-commerce web application.....	2
Carousel on the home page for the featured products.....	2
Fancy floating anchor on the home page to help with navigation	2
Offer on the shipping details page that automatically gives free shipping for totals over \$600.....	3
Chosen fourth functional element - product searching and category filtering on the shop page.....	3
Task 2b - Justify your choice of JavaScript approach to resolving the data validation. .	3
Task 3 - Implement forms with data validation and error handling.	4
Task 3a - Rationale for use of selected tools, technologies, frameworks and libraries	4
GitHub repository	7
Image Sources and Licensing.....	7

Task 2 - Use JavaScript to create a range of interactive elements.

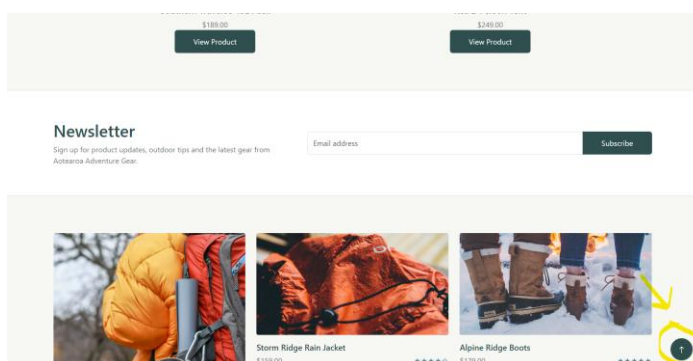
Task 2a - You must include at least four elements of functionality that are appropriate for an e-commerce web application.

I used JavaScript to add several interactive features throughout the website. On the home page, I created a featured product carousel that changes the products shown and responds to different screen sizes. I also added a floating “Back to Top” button that appears once the user scrolls down the page. On the shop page, I added product searching and category filtering so users can quickly find products. I also used JavaScript in the shopping cart and checkout process to update quantities, calculate totals, and automatically apply free shipping when the order total is over \$600.

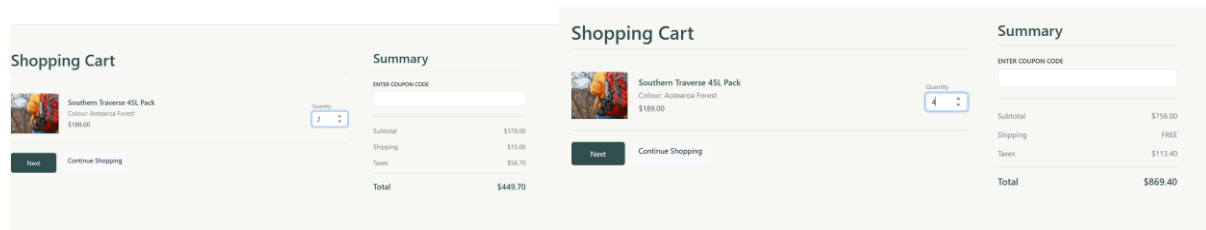
Carousel on the home page for the featured products



Fancy floating anchor on the home page to help with navigation

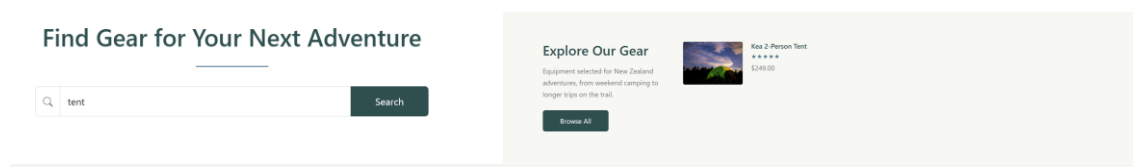


Offer on the shipping details page that automatically gives free shipping for totals over \$600.

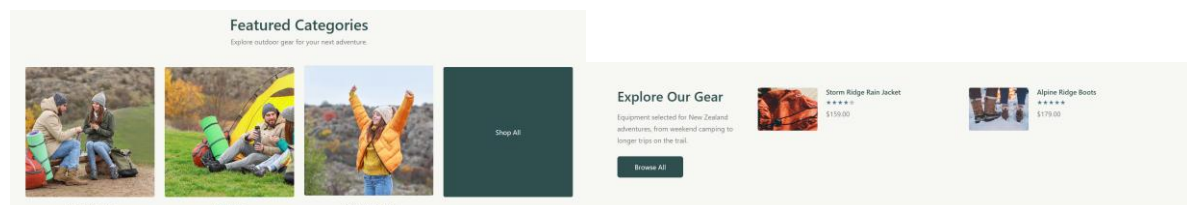


Chosen fourth functional element - product searching and category filtering on the shop page

The 'Explore our gear' section changes based on the interaction with the category buttons and the search bar.



Above you can see that the only item showing is a tent, based on my search.



After pressing the clothing category, only clothing items show.

Task 2b - Justify your choice of JavaScript approach to resolving the data validation.

I chose to use native JavaScript for the data validation and interactive features in this project. I did not use a JavaScript framework such as jQuery because the functionality required for the website was reasonably straightforward, and native JavaScript allowed me to meet the requirements without adding unnecessary libraries or dependencies. It

also gave me more practice working directly with the DOM, event listeners and JavaScript functions.

HTML5 validation was used where possible, such as required fields, email input types, minimum and maximum values, and other input restrictions. This provides a useful first level of validation because the browser can identify basic problems before the form is submitted. However, HTML5 validation was not enough for some of the more specific business rules required by the website.

For example, the shipping form checks that a New Zealand postcode contains exactly four digits and that the phone number follows an acceptable format. JavaScript allowed me to apply these specific rules and provide feedback when the input was invalid. In the shopping cart, JavaScript prevents invalid product quantities and dynamically recalculates the subtotal, tax, shipping and total when the quantity changes. This is important because these values depend on each other and cannot be handled by HTML5 validation alone.

On the payment page, JavaScript provides more specific checks for the 16-digit card number, MM / YY expiry format and 3–4 digit CVV. JavaScript also restricts inappropriate characters and formats some of the input while it is being entered. This provides immediate feedback and reduces the chance of invalid data reaching the final validation.

I also used localStorage to maintain the cart information between pages without requiring a database or server. Bootstrap was used as the UI framework for the responsive layout and components, while my native JavaScript was kept separately in custom.js for the custom behaviour and validation.

Task 3 - Implement forms with data validation and error handling.

Task 3a - Rationale for use of selected tools, technologies, frameworks and libraries

I used a combination of HTML5 validation and native JavaScript to validate the forms in the Aotearoa Adventure Gear website. HTML5 was used for basic validation where possible, including required fields, type="email", type="tel", number inputs, and minimum and maximum values. This allowed the browser to complete some of the

initial checks and helped prevent incomplete or obviously invalid information from being submitted.

Native JavaScript was then used where the website needed more specific validation rules. For example, on the shipping page JavaScript checks that a New Zealand postcode contains exactly four digits and uses a regular expression to check that the phone number only contains appropriate characters and is within an acceptable length. If the information is invalid, the user is given feedback and the form does not continue to the payment page until the problem has been corrected.

In the shopping cart, JavaScript prevents the product quantity from going below 1 or above 10. It also recalculates the subtotal, tax, shipping and total whenever the quantity changes. This helps prevent invalid quantities or calculated values from being carried through the checkout. The free-shipping rule is also calculated from the order value rather than relying on a value entered by the user.

I added more detailed validation to the payment page. The card number and CVV inputs remove non-numeric characters as the user types. This prevents letters and other inappropriate characters from being entered and reduces the chance of invalid data reaching the final validation. The card number is limited to 16 digits and the CVV to 3 or 4 digits.

JavaScript also formats the card number into groups of four digits and automatically formats the expiry date as MM / YY. This improves usability because the values are easier for the user to read and compare with their card. It also makes the expected format clearer and gives the validation code a consistent value to check. Regular expressions are then used to check the completed card number, expiry and CVV before the form is accepted. The expiry validation also checks that the month is between 01 and 12.

My validation uses a layered approach. HTML5 handles suitable basic validation first, JavaScript applies the more specific rules required by the application, and invalid data prevents the user from progressing through the checkout. This makes the forms more robust while also providing immediate feedback to the user. However, this is client-side validation and could be bypassed. In a real e-commerce application, the same data would also need to be validated securely on the server before an order or payment was processed.

I chose native JavaScript rather than a framework such as jQuery because the validation and interaction required for this project could be achieved without adding another dependency. Bootstrap was used for the responsive layout and UI components, while my custom validation and interactive functionality were kept in custom.js. Keeping this logic together made the JavaScript easier to organise, maintain and reuse across the different pages.



Southern Traverse 45L Pack

Colour: Aotearoa Forest

\$189.00

Quantity

Next

Continue Shopping

ENTER COUPON CODE

Subtotal\$189.00

Shipping\$15.00

Taxes\$28.35

Total\$232.35

Only allows valid quantaties in the cart.

Shipping Details

First Name

Last Name

Address

Please fill in this field.

Address 2 (optional)

Country

City

Postcode

Phone Number

Standard Shipping

2-5 working days

Express Delivery

1-2 working days

Next

Back

Shipping Details

Molly

Smith

19 buccaneer court

Address 2 (optional)

New Zealand

Auckland

sdhff

sdhuff

Standard Shipping

2-5 working days

Express Delivery

1-2 working days

Back

Summary



Southern Traverse 45L Pack

Aotearoa Forest

Quantity: 1

HAVE A RECEIPT?

Subtotal\$189.00

Shipping\$15.00

Taxes\$28.35

Total\$232.35

Shipping Details

Molly

Smith

19 buccaneer court

Address 2 (optional)

New Zealand

Auckland

322222

sdhuff

Standard Shipping

2-5 working days

Express Delivery

1-2 working days

Back

This page says

Please enter a valid phone number.

OK

Shipping Details

Smith

Address 2 (optional)

Auckland

sdhuff

Express Delivery

1-2 working days

Examples of form validation on the shipping details page.

Credit Card

1231 2312 3123 1231

11 / 29

CVV

Cardholder Name

PayPal

Continue using PayPal as the selected payment method. No real payment will be processed.

Pay Now

Back

Automatic formatting for the Card number and expiry date

The image displays three screenshots of a payment form titled "Payment Method".

- Top Left Screenshot:** Shows the "Credit Card" section with a card number field containing "0000 0000 0000 0000", an expiry date field with "MM / YY", and a CVV field. A tooltip message "Please fill in this field." is visible over the card number field.
- Top Right Screenshot:** Shows the "Credit Card" section with a card number field containing "3333", an expiry date field with "33 / 33", and a CVV field containing "333". A tooltip message "Please enter a valid 16-digit card number." is visible.
- Bottom Screenshot:** Shows the "Credit Card" section with a card number field containing "3333 1111 1111 1111", an expiry date field with "22", and a CVV field containing "333". A tooltip message "Please enter the expiry date in MM / YY format." is visible.

Each screenshot also shows the "PayPal" section with a note: "Continue using PayPal as the selected payment method. No real payment will be processed." and buttons for "Pay Now" and "Back".

Further example of form validation on the payment page.

GitHub repository

<https://github.com/molly-smith-x/AS2>

Image Sources and Licensing

All photographic images used on the Aotearoa Adventure Gear website were sourced from StockSnap. I specifically selected images that were available under the **Creative Commons CC0 licence**. This allows the images to be copied, modified and used for personal or commercial purposes without requiring attribution.

Although attribution is not required, I have included the original image sources below to document where the website assets came from.

- [People Hiking – StockSnap](#)
- [Couple Tent – StockSnap](#)

- [Guy Man – StockSnap](#)
- [Hiker Backpack – StockSnap](#)
- [Red Jacket – StockSnap](#)
- [Snow Winter – StockSnap](#)
- [Night Tent – StockSnap](#)
- [Hiking Couple – StockSnap](#)
- [Scenic Snowy – StockSnap](#)
- [Hiking Fall – StockSnap](#)
- [Woman Happy – StockSnap](#)
- [Wood Outdoor – StockSnap](#)
- [Grill Camp – StockSnap](#)